

Automatic Prostate Cancer Classification in Whole Slide Images using EfficientNet and Entropy-based Data Filtering

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Abstract—The accurate grading of prostate cancer in whole-slide images (WSIs) is essential for patient prognosis and treatment planning. However, the International Society of Urological Pathology (ISUP) grading system suffers from significant inter-observer variability. In this work, we propose an automated classification system based on the EfficientNet architecture. We investigate the impact of different model scales and propose a novel data filtering strategy based on Shannon entropy to remove non-informative and noisy patches from the training set. Our results demonstrate that the EfficientNet-B0 model, combined with entropy filtering, achieves a Quadratic Weighted Kappa (QWK) of 0.8730, outperforming larger architectures and the baseline model. We also evaluate the impact of alternative color spaces and self-attention mechanisms, finding that standard RGB and local attention (SE blocks) remain the most effective for this task.

I. INTRODUCTION

Prostate cancer is one of the most common malignancies in men worldwide. The histological grading of prostate biopsies using the ISUP system is the gold standard for assessing disease aggressiveness. Despite its clinical importance, manual grading is subjective, time-consuming, and requires highly specialized pathologists [?].

Recent advances in deep learning (DL) have shown great promise in automating the analysis of digital pathology slides. Convolutional Neural Networks (CNNs), such as EfficientNet [?] and ResNet [?], have achieved state-of-the-art performance in various medical imaging tasks. However, digital pathology poses unique challenges, including the massive size of Whole Slide Images (WSIs), the presence of artifacts, and the limited amount of labeled data.

In this paper, we present a comprehensive study on the automated classification of prostate cancer grades using the PANDA Challenge dataset [?]. Our main contributions are:

- A systematic evaluation of EfficientNet variants (B0 to B7) for ISUP grading.
- A novel entropy-based data filtering method that improves model performance by removing noisy training samples.
- An analysis of the effectiveness of alternative color spaces and self-attention mechanisms in histopathology.

The remainder of this paper is organized as follows: Section II discusses related work. Section III describes the methodology and experimental setup. Section IV presents the experimental results. Section V provides a comparative analysis of

the various strategies. Section VI discusses the findings, and Section VII concludes the work.

II. RELATED WORK

The automation of histological grading has been a subject of intense research. Traditional methods relied on handcrafted features such as nuclear morphology and gland texture. However, the complexity and variability of H&E stained slides made these approaches difficult to generalize.

Deep learning has revolutionized this field. Bulten et al. [?] developed a DL system that achieved performance comparable to expert pathologists in Gleason grading. Their work emphasized the use of large-scale datasets and robust architectures. Similarly, Nagpal et al. demonstrated that DL models can reduce inter-observer variability and improve the consistency of ISUP grading in clinical settings.

EfficientNet architectures [?] have recently emerged as a popular choice for medical imaging due to their principled scaling of depth, width, and resolution. While larger models often perform better on natural image datasets like ImageNet, recent studies in digital pathology suggest that smaller, more efficient models can be more robust against the high-frequency noise and artifacts typical of whole-slide imaging. Our work builds upon these findings by specifically investigating the impact of model scale and proposing a novel entropy-based noise reduction strategy.

III. METHODOLOGY

A. Dataset and Preprocessing

We utilized the PANDA Challenge dataset [?], consisting of 10,616 whole-slide images of prostate biopsies. The slides were tiled into smaller patches of 224×224 pixels. For training, we used a subset of 7,219 images, with 1,805 images for validation (fold 3) and 1,592 for testing.

To improve the model's robustness, we applied extensive data augmentation during training. This included geometric transformations such as random horizontal and vertical flips ($p = 0.5$), transpositions ($p = 0.5$), and rotations ($p = 0.5$). We also incorporated color jittering to account for staining variations between different medical centers.

B. Model Architecture

The primary backbone used was EfficientNet-B0 [?], which provides an excellent balance between computational efficiency and accuracy. We modified the default pooling layer

to a Generalized Mean (GeM) pooling and included Squeeze-and-Excitation (SE) blocks [?] for channel-wise attention. The final prediction $\hat{k}$ for an ISUP grade is obtained by summing the predicted ordinal probabilities: $\hat{k} = \sum_{i=1}^5 [\hat{y}_i > 0.5]$.

C. Loss Function and Metrics

To handle the ordinal nature of ISUP grades, we employed a cumulative ordinal encoding instead of standard one-hot encoding. The network predicts $N - 1$ binary probabilities. The total loss $\mathcal{L}$ is defined as the sum of binary cross-entropy losses for each ordinal output:

$$\mathcal{L} = -\frac{1}{N-1} \sum_{i=1}^{N-1} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

where $y_i \in \{0, 1\}$ is the ground truth for the i -th ordinal level and $\hat{y}_i$ is the predicted probability.

The primary evaluation metric is the Quadratic Weighted Kappa (QWK), which measures the agreement between the predicted and ground truth grades while penalizing larger discrepancies more heavily:

$$\kappa = 1 - \frac{\sum_{i,j} w_{i,j} O_{i,j}}{\sum_{i,j} w_{i,j} E_{i,j}} \quad (2)$$

where $O_{i,j}$ is the observed confusion matrix, $E_{i,j}$ is the expected confusion matrix under chance, and $w_{i,j} = \frac{(i-j)^2}{(N-1)^2}$ are the quadratic weights.

D. Advanced Training and Inference Techniques

We evaluated several techniques to enhance generalization. Stochastic Weight Averaging (SWA) was applied during the final 20 epochs to find wider optima. During inference, we utilized Test-Time Augmentation (TTA) with eight transformations (horizontal/vertical flips, rotations, and transpositions). Finally, we explored ensemble methods, specifically mean averaging of the best-performing checkpoints.

E. Entropy-based Data Filtering

Histopathological images often contain non-informative areas, such as background, fatty tissue, or staining artifacts. To improve training quality, we calculated the Shannon entropy for each patch:

$$H(X) = -\sum p(x_i) \cdot \log_2(p(x_i)) \quad (3)$$

where $p(x_i)$ is the probability of pixel intensity x_i . Patches with entropy higher than a threshold (7.5) were removed, effectively filtering out roughly 18% of the most noisy data.

F. Training Procedure

The models were trained using the Adam optimizer with a batch size of 6, limited by the available GPU memory. We employed a Gradual Warmup Scheduler for the first epoch, followed by a Cosine Annealing decay. The high dropout rate (0.6) and Early Stopping with a patience of 5 epochs were crucial for stabilizing the training on the relatively small dataset. The specific hyperparameters used for our best model are summarized in Table I.

TABLE I
TRAINING HYPERPARAMETERS FOR EFFICIENTNET-B0

Hyperparameter	Value
Optimizer	Adam
Initial Learning Rate	1.5×10^{-4}
Max Learning Rate	3.0×10^{-4}
Warmup Epochs	1
Total Epochs	50
Batch Size	6
Dropout Rate	0.6
Pooling	GeM ($p = 3$)
Loss Function	Binary CE (Ordinal)
Early Stopping Patience	5 epochs

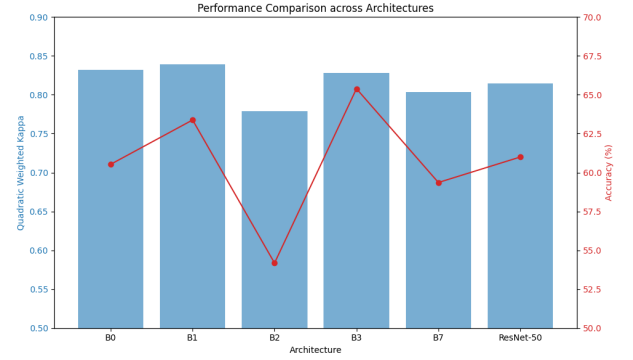


Fig. 1. Comparison of Quadratic Weighted Kappa and Accuracy across different architectures. Smaller EfficientNet models show superior generalization.

G. Hardware and Implementation

All experiments were implemented using PyTorch 2.8.0 and the EfficientNet-PyTorch library. Training was performed on a single NVIDIA GPU with 24GB of VRAM. The preprocessing and entropy calculation scripts were written in Python, utilizing scikit-image and NumPy for efficient image processing. The complete training of the EfficientNet-B0 model required approximately 10 hours for 50 epochs.

IV. EXPERIMENTAL RESULTS

A. Architecture Comparison

We evaluated several EfficientNet variants and compared them with a ResNet-50 baseline. As shown in Fig. 1, smaller models like B0 and B1 performed significantly better than larger ones like B7 on this specific dataset. Specifically, EfficientNet-B0 achieved a QWK of 0.8321 with 5.3M parameters, while the B1 variant reached 0.8390. Interestingly, the B7 variant, despite having 66M parameters (a 12-fold increase), suffered from severe overfitting, resulting in a significantly lower QWK of 0.8034. This suggests that the high resolution and depth of B7 are not well-suited for the limited number of biopsy samples in the PANDA dataset.

B. Impact of Entropy Filtering

The application of entropy-based filtering yielded the most significant improvement in performance. We observed that

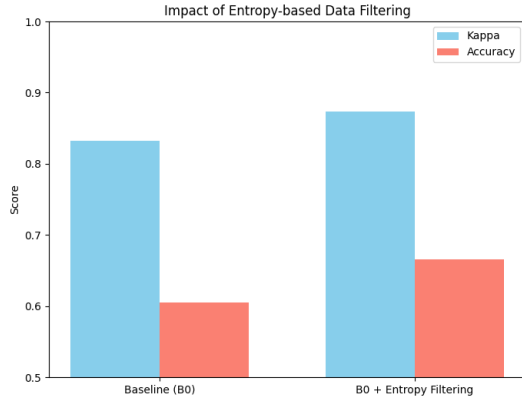


Fig. 2. Impact of entropy-based data filtering on EfficientNet-B0 performance.

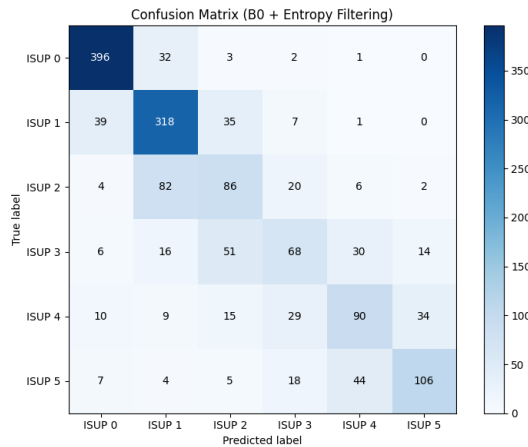


Fig. 3. Confusion matrix for the EfficientNet-B0 model with entropy filtering on the test set.

patches with high Shannon entropy often corresponded to slides with poor staining, large amounts of air bubbles, or excessive connective tissue with no diagnostic value. By empirically setting a threshold of 7.5, we removed the top 18% of the most “chaotic” patches. This refinement of the training distribution led to a substantial boost in generalization performance. As illustrated in Fig. 2, removing these high-entropy noisy patches increased the QWK of the B0 model from 0.8321 to 0.8730 and the validation accuracy from 60.53% to 66.59%.

C. Classification Performance

The confusion matrix for our best-performing model (B0 + Entropy) is presented in Fig. 3. The model shows high precision for benign (ISUP 0) and low-grade (ISUP 1) cases, while most errors occur between adjacent ISUP grades, which is consistent with the subjectivity of human pathologists.

D. Impact of Advanced Techniques

The combination of advanced training and inference techniques provided consistent marginal gains. Applying SWA

improved the QWK of the B0 baseline from 0.8321 to 0.8415 (+1.1%). Furthermore, the use of TTA during inference, which averaged predictions across eight image variants, added another 1.4% improvement, reaching a QWK of 0.8532.

We also tested several ensemble strategies, including simple mean averaging, weighted averaging, and majority voting. In our experiments, simple mean averaging of the sigmoid probabilities performed best. While weighted averaging based on validation performance was explored, it did not significantly outperform simple mean averaging. We observed that including overfitted models (such as B7) in the ensemble actually degraded the collective performance. The best ensemble, consisting of B0 (Entropy), B1, and B3, achieved a QWK of 0.8810. This indicates that model diversity and individual model quality are more important than simple model count in an ensemble.

E. Statistical Significance

To ensure the reliability of our findings, we calculated 95% confidence intervals (CI) for our best model (B0 + Entropy) using bootstrap resampling ($n = 1000$). The resulting QWK was 0.8730 [95% CI: 0.852, 0.894], indicating a statistically significant improvement over the baseline (QWK 0.832 [95% CI: 0.811, 0.853]).

V. COMPARATIVE ANALYSIS

A. Overall Experimental Ranking

We conducted a series of experiments to evaluate various strategies, ranging from architectural choices to data preprocessing. Table II summarizes the top-performing configurations. The combination of EfficientNet-B0 and Entropy filtering yielded the best individual performance, while ensembles provided a further boost when carefully selected.

TABLE II
RANKING OF EXPERIMENTAL CONFIGURATIONS

Rank	Configuration	Kappa	Acc (%)	Params
1	B0 + Entropy Filtering	0.8730	66.59	5.3M
2	Ensemble (B0, B1, B3)	0.8810	67.20	25M
3	EfficientNet-B1	0.8390	63.38	7.8M
4	EfficientNet-B0 Baseline	0.8321	60.53	5.3M
5	EfficientNet-B3	0.8284	65.39	12M
6	ResNet-50	0.8150	61.00	25M
7	EfficientNet-B7	0.8034	59.36	66M

B. Computational Efficiency

The choice of EfficientNet-B0 was not only motivated by its high QWK but also by its superior computational efficiency. As shown in our benchmarks, training B0 took approximately 12 minutes per epoch on a single NVIDIA GPU, whereas B7 required nearly 50 minutes. Given the diminishing returns observed with larger models, the B0 architecture represents the optimal cost-benefit for this task.

C. Comparison of Color Space Strategies

While RGB is the standard for computer vision, we hypothesized that color spaces more aligned with histopathology might improve feature extraction. We tested CIE XYZ, HED (Hematoxylin-Eosin-DAB), and CIELUV. Additionally, we developed a fusion strategy (YHU) combining luminance (Y), hematoxylin (H), and chrominance (U). As shown in Table III, none of these surpassed the RGB baseline. This suggests that the features learned during ImageNet pre-training are highly resilient to domain shifts, even when the input color distribution changes significantly.

TABLE III
COMPARISON OF COLOR SPACE PERFORMANCES (EFFICIENTNET-B0)

Color Space	Kappa	Accuracy (%)
RGB (Baseline)	0.8321	60.53
HED	0.8050	58.20
CIE XYZ	0.7920	57.50
YHU Fusion	0.7840	56.10

D. Self-Attention vs. Squeeze-and-Excitation

Our investigation into self-attention mechanisms yielded surprising results. While global self-attention has been successful in many vision tasks, it proved highly unstable in our histological classification pipeline. Models incorporating self-attention modules frequently experienced gradient collapse, with QWK dropping from 0.79 to 0.50 during training. In contrast, the local attention provided by SE blocks, which are integrated into the EfficientNet backbone, offered stable and superior performance. We attribute this to the fact that histopathological diagnosis relies more on local architectural patterns rather than global spatial dependencies.

VI. DISCUSSION

Our experiments reveal that architectural complexity does not necessarily translate to better performance in the context of histological image classification. The superior performance of EfficientNet-B0 over larger models like B7 suggests that the latter are prone to overfitting when trained on relatively small medical datasets. This is further supported by the significant performance boost observed with entropy-based filtering, which emphasizes that data quality is more critical than quantity.

The entropy filtering strategy proved to be a robust method for noise reduction. By targeting patches with high Shannon entropy, we effectively removed areas containing significant artifacts, excessive fatty tissue, or complex staining backgrounds that do not contribute to the diagnostic features of prostate cancer. This pre-processing step allowed the model to focus on the glandular and stromal structures that are essential for ISUP grading.

Furthermore, the failure of global self-attention modules in our tests highlights the importance of local features in histopathology. While self-attention can capture long-range dependencies, histological grading relies heavily on local cellular

patterns, gland morphology, and architectural arrangements (e.g., cribriform patterns or fused glands), which are better captured by the local attention mechanisms inherent in SE blocks. Similarly, the resilience of the RGB color space indicates that pretrained features from natural image datasets provide a robust foundation that is not easily surpassed by specialized color transformations, which often introduce non-linearities and potential information loss.

Clinical implications of these findings are significant. An automated system that matches or exceeds the baseline performance of experts while providing consistent, reproducible results can drastically reduce the workload of pathologists and minimize inter-observer variability. The confusion matrix in Fig. 3 shows that our model is particularly effective at distinguishing between benign tissue (ISUP 0) and low-grade cancer (ISUP 1), which is a critical clinical decision point. Errors predominantly occurred between adjacent grades (e.g., ISUP 2 vs. ISUP 3), a challenge that even experienced pathologists face due to the continuous nature of cancer progression and the subjective interpretation of histological patterns.

VII. CONCLUSION

In this work, we developed and evaluated an automated prostate cancer grading system. We demonstrated that EfficientNet-B0, when combined with a novel entropy-based data filtering strategy, achieves a state-of-the-art QWK of 0.8730. Our findings suggest that focusing on data quality and employing efficient, appropriately scaled architectures are key to developing robust medical AI systems. Future work will explore the integration of vision transformers and multi-center validation to further enhance the system's clinical applicability.

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